

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				Alternative 1 Weekly mass of cow dung = 9 000 [kg] (1) Yearly mass of cow dung = 468 000 (ecf) (1) Gas/year = 468 000 (ecf) $\times$ 0.05 = 23 400 (1) Units/year 23 400 (ecf) $\times$ 1.9 (1) = 44 460 (ecf) [kWh] (1) [so incorrect] Alternative 2 Weekly gas/cow = 60 $\times$ 0.05 = 3 (1) Weekly units/cow = 3 (ecf) $\times$ 1.9 = 5.7 (1) Weekly units for herd of cows = 5.7(ecf) $\times$ 150 = 855 (1) Units/year = 855 (ecf) $\times$ 52 (1) = 44 460 (ecf) [kWh] (1) [so incorrect] Alternative 3 Weekly units required = $\frac{144\,000}{52} = 2\,769.23$ (1) Weekly volume of gas required = $\frac{2\,769.23\text{ ecf}}{1.9} = 1\,457.5$ (1) Weekly mass of dung required = $\frac{1\,457.5\text{ ecf}}{0.05} = 29\,149.8$ (1) Number of cows required = $\frac{29\,149.8\text{ ecf}}{60}$ (1) = 486 (ecf) (1) [so incorrect] Accept for 4 marks a final answer of 148 200 [kWh] Common answer 235 638 award 4 marks or 296.4 award 4 marks also or 124 020						